Evaluating Multi-modal Language Models Through Concept Hacking

Yijiang Li[1], Bingyang Wang[2], Tianwei Zhao[3], Qingying Gao[3], Hokin Deng[4], Dezhi Luo[5,6]

[1]UCSD, [2]Emory, [3]JHU, [4]CMU, [5]UMich, [6]UCL

ICLR

Introduction

Evaluating the cognitive abilities of Multi-modal Language Models (MLLMs) is challenging due to their reliance on spurious correlations. To distinguish shortcut-taking from genuine reasoning, we introduce **Concept Hacking, a paradigm manipulating concept-relevant information to flip the ground-truth but preserving concept-irrelevant confounds**. For instance, in a perceptual constancy test, models must recognize that a uniformly wide bridge does not narrow in the distance; the manipulated condition using concept hacking altered the bridge to actually taper.

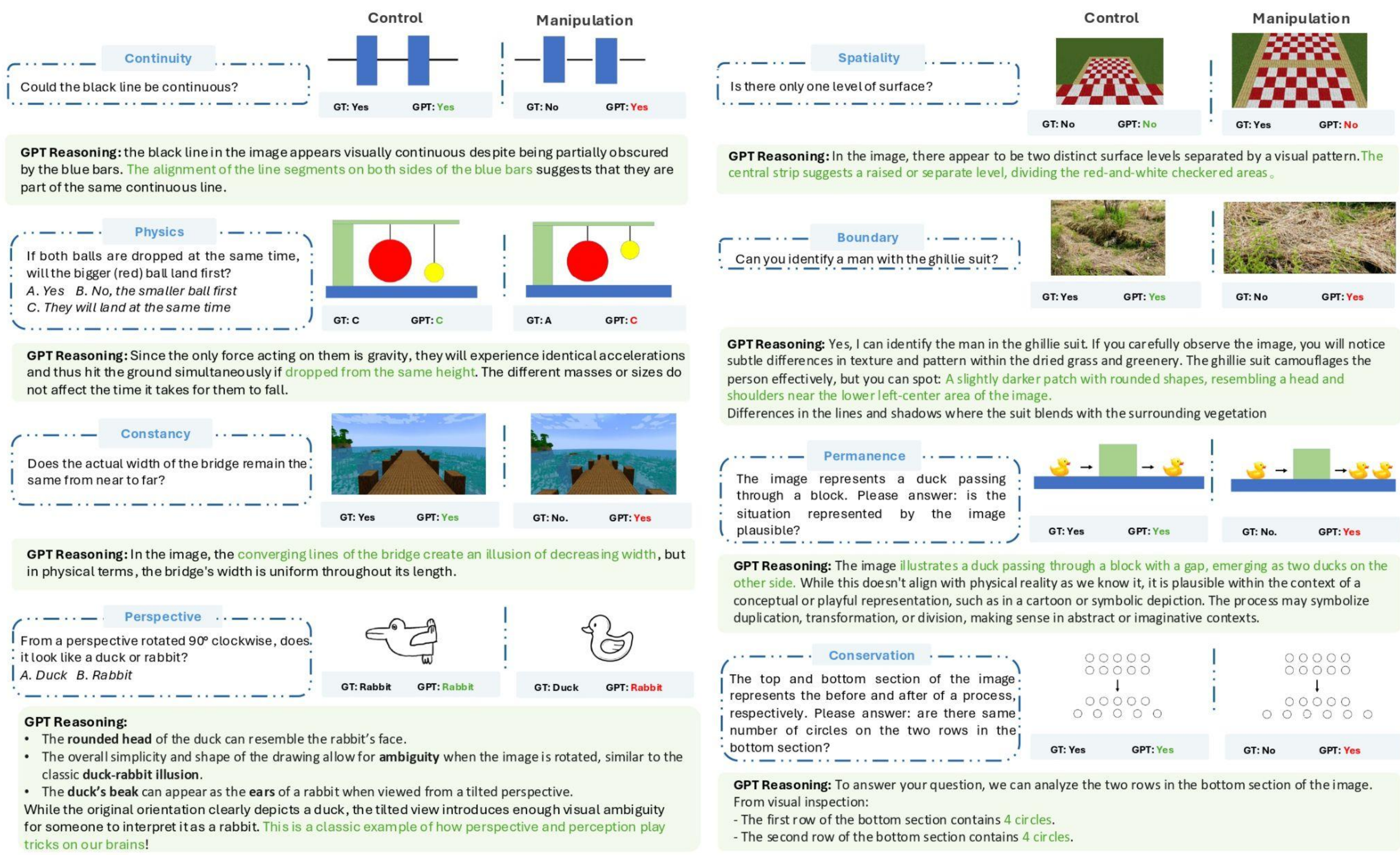
We assessed **209 models** across **45 experiment** pairs spanning nine low-level cognitive abilities, encompassing all five core knowledge domains. Comparing performance on manipulated versus standard conditions revealed that models fell into **shortcut-reliant** or **illusory understanding** types, with **none approaching human-level performance**. Models of varying sizes appear in each category, indicating that scaling neither imparts core knowledge nor reduces shortcut reliance. These findings highlight fundamental limitations in current MLLMs, reinforcing concerns about their ability to achieve genuine understanding.

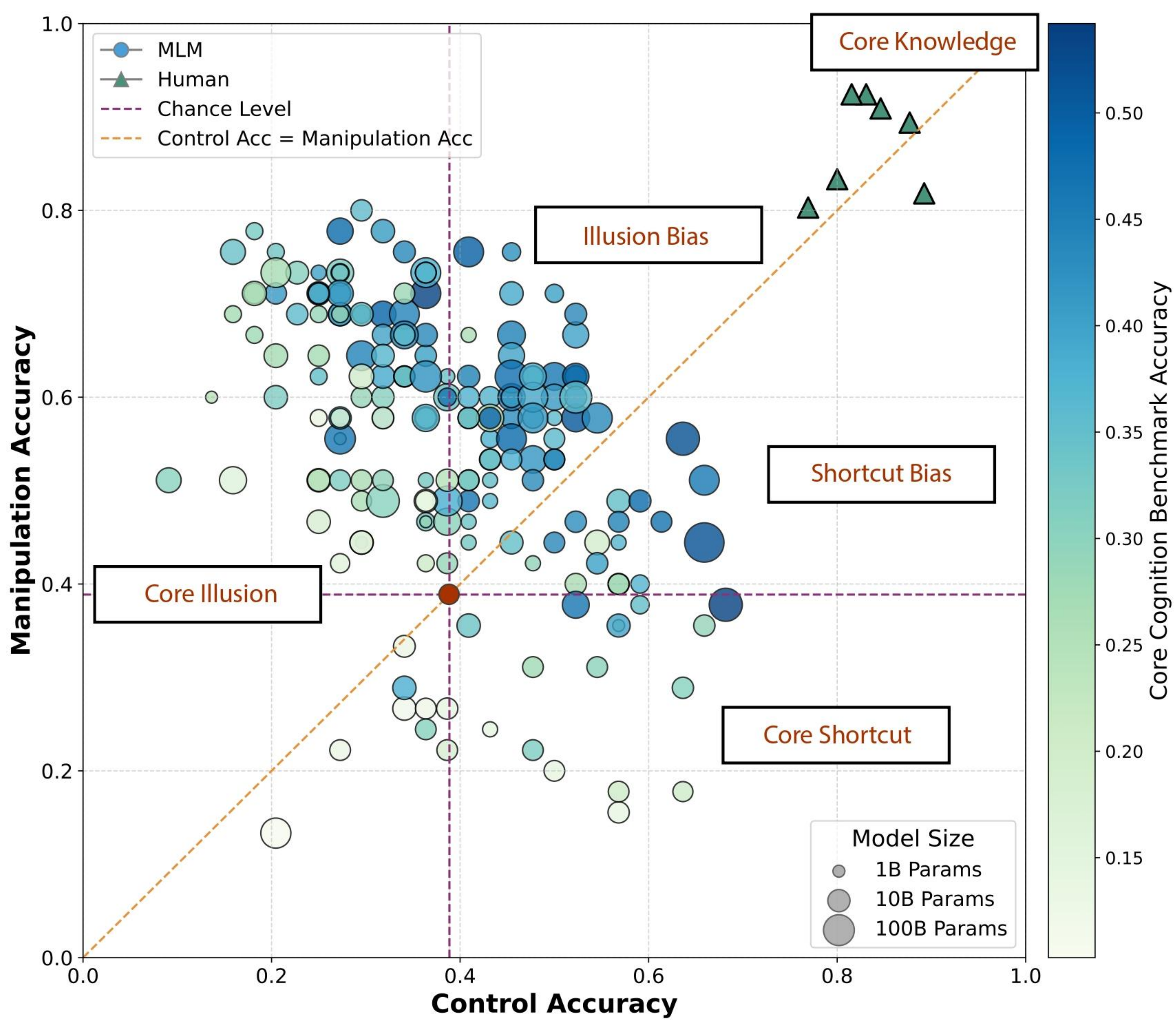
Methods

Concept hacking systematically manipulates task-relevant details in core knowledge assessments to completely invert the ground truth while preserving all task-irrelevant conditions.

We applied concept hacking to 45 standard tasks, resulted in a total of 90 tasks (45 manipulated and 45 corresponding standard control tasks), each five pairs probe one of the nine low-level abilities.

The comparison between an individual's performance on a manipulation task and their corresponding control is capable of revealing three distinctive strategies: core knowledge, shortcut-taking (solves control but not manipulation), and illusory understanding (cannot solve control).





Proprietary Models				Open Source Models			
Model Name	Control Accuracy	Manipulation Accuracy	Serie	*Model Name*	Control Accuracy	Manipulation Accuracy	Serie
claude-3-5-sonnet	44.44%	62.22%	claude3_5	llava_next_110b	35.56%	62.22%	llava_next
claude-3-sonnet	31.11%	48.89%	claude3	llava_next_72b	46.67%	57.78%	llava_next
claude-3-opus	53.33%	60.00%	claude3	llava_next_mistral_7b	44.44%	53.33%	llava_next
claude-3-haiku	20.00%	73.33%	claude3	llava_next_llama3	40.00%	60.00%	llava_next
gemini-1.5-pro	64.44%	55.56%	gemini	Qwen2.5-VL-3B-Instruct	33.33%	62.22%	qwen2_5_vl
gemini-1.5-flash	53.33%	37.78%	gemini	Qwen2-VL-72B-Instruct	66.67%	51.11%	qwen2vl
gemini-1.5-flash-8b	55.56%	42.22%	gemini	Qwen2-VL-7B-Instruct	48.89%	44.44%	qwen2vl
gpt-4-turbo	66.67%	44.44%	gpt	Qwen2-VL-2B-Instruct	35.56%	51.11%	qwen2vl
gpt-4o	68.89%	37.78%	gpt	mPLUG-Owl3	62.22%	46.67%	mplug3
reka-core	31.11%	60.00%	reka	NVLM-D-72B	40.00%	75.56%	nvlm
reka-flash	26.67%	62.22%	reka	deepseek-vl2	26.67%	71.11%	deepseek2
reka-edge	22.22%	64.44%	reka	deepseek-vl2-small	17.78%	71.11%	deepseek2
qwen-vl-plus	64.44%	44.44%	qwen_vl	deepseek-vl2-tiny	17.78%	66.67%	deepseek2
qwen-vl-max	57.78%	60.00%	qwen_vl	360VL-70B	28.89%	64.44%	360vl-series

Model Performances

Results demonstrated a clear segregation of models relying on shortcut-taking and illusory understanding. Larger models did not perform better on the benchmark, even comparing to smaller models from the same series. Our findings support the hypothesis that MLLMs lack core knowledge.

Project
Growing AI Like A Child
Website:

Main Paper
Core Knowledge Deficits in MLLMs
Preprint: